Week 05 Research Report: Minimax Game Tree Search, SearchSpaceMode Distinction, and Horizon Effect Analysis

Project Title: Research, Design, and Implementation of an Optimized Gomoku Game Engine with Artificial Intelligence **Academic Module:** Basic Project (■■ Án C■ S■) — Software Engineering & AI Track **Document Standard:** Standard 00 (IMRaD & Evidence Specification for Software Systems) **Control Layer Reference:** `RQ-W5-01`, `EXP-W5-01`, `CFG-SEARCH-v1` **Authoritative Repository:** [GitHub – tducn110/caro](https://github.com/tducn110/caro) (commit `cb6fd6f`) **Language:** English **Status:** Complete & Verified

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1. Plan & Research Questions

1.1 Problem Formulation

Heuristic static evaluation functions evaluate only the immediate board state ($d = 0$), leaving the engine vulnerable to two-ply tactics such as fork creation (simultaneous open-threes) or delayed counter-threats. Lookahead game tree search is mandatory. However, unpruned Minimax suffers from combinatorial explosion: evaluating a tree of depth d with branching factor b requires expanding $\mathcal{O}(b^d)$ nodes.

Crucially, in scientific benchmarking, **Candidate Move Generation is a search-space restriction, NOT part of pure Minimax**. If candidate generation is implicitly bundled into the Minimax baseline, the baseline and the optimization become conflated, distorting the empirical evidence. Therefore, the search space must be formally parameterized via `SearchSpaceMode`.

1.2 Core Research Questions

1. **RQ-W5-01 (SearchSpaceMode Divergence):** What are the empirical node expansion curves and computational limits of pure Minimax under `FULL_LEGAL` ($A(s)$) versus `CANDIDATE_RESTRICTED` ($C(s) \subseteq A(s)$)?
2. **RQ-W5-02 (Branching Compression):** What is the exact empirical branching factor b_{eff} under `FULL_LEGAL` ($b \approx 200$) compared to `CFG-CAND-v1` ($b_{\text{eff}} \approx 18$)?
3. **RQ-W5-03 (Horizon Vulnerability):** How does fixed search depth induce the "Horizon Effect", where tactical threats immediately beyond depth d remain undetected until depth $d+1$ or $d+2$?

```
graph TD
    Root["Root State (Max Player)"] --> ModeChoice{"SearchSpaceMode"}

    ModeChoice -- "FULL_LEGAL A(s)" --> FullSearch["Branching Factor b0 ≈ 200\nDepth 2: 40,000 nodes\nDe
    ModeChoice -- "CANDIDATE_RESTRICTED C(s)" --> CandSearch["Branching Factor b_eff ≈ 18 (r <= 2)\nDept

    FullSearch --> Backprop["Minimax Backpropagation (Min / Max)"]
    CandSearch --> Backprop
    Backprop --> BestMove["Optimal Move Decision"]
```

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2. Background & Literature Citations

- **[B] Von Neumann, J. (1928).** *Zur Theorie der Gesellschaftsspiele.* Mathematische Annalen, 100(1), 295–320. Formalized the Minimax theorem for zero-sum, two-player games of perfect information.
- **[B] Shannon, C. E. (1950).** *Programming a Computer for Playing Chess.* Philosophical Magazine, 41(314), 256–275. Introduced depth-limited search and terminal leaf evaluation.
- **[B] Berliner, H. (1973).** *Some Necessary Conditions for a Master-Level Chess Program.* Computers and People, 22(8), 12–15. Formally defined the "Horizon Effect" in lookahead tree search.

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3. Methods & Search Space Specification

3.1 Formal Definition of SearchSpaceMode

To isolate the algorithmic effect of candidate pruning from search tree mechanics, all search routines must explicitly declare their search space mode:

$$\text{SearchSpaceMode} \in \{ \text{FULL_LEGAL}, \text{CANDIDATE_RESTRICTED} \}$$

1. `FULL_LEGAL`**:** The move generator returns the entire set of unoccupied board cells:

$$A(s) = \{ c \in [0, 224] \mid \text{cells}[c] = \text{Empty} \}$$ *Properties:* Guaranteed game-tree completeness; strictly preserves mathematical Minimax semantics.

1. `CANDIDATE_RESTRICTED` **(**`CFG-CAND-v1`**):** The move generator restricts search to empty cells within Chebyshev distance $r \le 2$ of any currently occupied stone:

$$C(s) = \{ c \in A(s) \mid \exists c' \text{ s.t. } \text{cells}[c'] \ne \text{Empty} \land \|(x_c, y_c) - (x_{c'}, y_{c'})\|_{\infty} \le 2 \}$$ *Properties:* Drastically compresses branching factor; introduces tactical recall risk if r is too small.

3.2 Standardized Experiment Naming Convention

All experimental baselines are designated using the formal configuration schema:

```
<ALGO>-<SPACE>-<CONFIG>-D<DEPTH>
```

- `MM-FULL-D2`: Minimax, Full Legal Search Space, Depth 2.
- `MM-CAND-v1-D2`: Minimax, Candidate-Restricted (`CFG-CAND-v1`), Depth 2.
- `MM-CAND-v1-D4`: Minimax, Candidate-Restricted (`CFG-CAND-v1`), Depth 4.
- `AB-FULL-D2`: Alpha-Beta, Full Legal Search Space, Depth 2.
- `AB-CAND-v1-D4`: Alpha-Beta, Candidate-Restricted (`CFG-CAND-v1`), Depth 4.

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4. Implementation & Code Contracts

4.1 Rust SearchSpaceMode Parameterization (`caro-core-reference / src/minimax.rs`)

```
#[derive(Copy, Clone, Debug, PartialEq, Eq)]
pub enum SearchSpaceMode {
    FullLegal,
    CandidateRestricted(isize), // Radius r
}

pub struct MinimaxConfig {
    pub depth: usize,
    pub space_mode: SearchSpaceMode,
    pub maximizing_player: Player,
}

pub fn minimax_search(
    board: &mut Board,
    config: &MinimaxConfig,
    depth: usize,
    is_maximizing: bool,
    node_count: &mut usize,
) -> i32 {
    *node_count += 1;

    // Terminal detection (Five in a row)
    if let Some(winner) = check_winner_fast(board) {
        return if winner == config.maximizing_player {
            SCORE_FIVE + (depth as i32 * 100)
        } else {
            -SCORE_FIVE - (depth as i32 * 100)
        };
    }

... [truncated for print length] ...
```

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5. Experimental Setup & Horizon Failure Corpus

5.1 Experiment Protocol (`EXP-W5-01`)

- **Independent Variables:** Search Depth $d \in \{1, 2, 3, 4\}$, `SearchSpaceMode` (`FullLegal` vs `CandidateRestricted`).
- **Controlled Variables:** `CFG-RULE-v1` (15×15 Gomoku), `CFG-EVAL-v1` (Calibrated weights), hardware runtime.
- **Dependent Variables:** `nodesVisited`, `leafEvaluations`, `searchTimeMs`, decision correctness.

5.2 Horizon Failure Corpus (`FIX-HORIZON`)

To scientifically prove the Horizon Effect, specific tactical positions were crafted where a threat matures at depth $d+1$:

- `FIX-HORIZON-001` **(Delayed 3-Ply Four-Three Fork):**
- Black has setup an Open Three and Broken Three.
- At $d = 2$, White AI does not foresee the simultaneous fork threat and plays an irrelevant move.
- At $d = 4$, White AI foresees the unstoppable four-three fork and plays the preemptive blocking move.
- `FIX-HORIZON-002` **(Over-the-Horizon Block):**
- Immediate win is not available, but failure to block an opponent diagonal extension at ply 3 leads to forced loss at ply 5.

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6. Results & Verification Evidence

6.1 EXP-W5-01: Full Legal vs. Candidate Restricted Scaling

Experiment Baseline	SearchSpace Mode	Depth (d)	Active Moves (b_{eff})	Nodes Visited (N_v)	Search Time (T_{search})	Feasibility Status
`MM-FULL-D1`	FullLegal ($A(s)$)	1	221	222	0.15 ms	Real-time OK
`MM-FULL-D2`	FullLegal ($A(s)$)	2	221	48,842	28.40 ms	Real-time OK
`MM-FULL-D3`	FullLegal ($A(s)$)	3	221	$\sim 1.07 \times 10^7$	$\sim 6,200\text{ ms}$	**TIMEOUT (> 1000ms)**
`MM-CAND-v1-D1`	Candidate ($r \le 2$)	1	18	19	$< 0.05\text{ ms}$	Real-time OK
`MM-CAND-v1-D2`	Candidate ($r \le 2$)	2	18	343	0.22 ms	Real-time OK
`MM-CAND-v1-D3`	Candidate ($r \le 2$)	3	18	6,175	3.84 ms	Real-time OK
`MM-CAND-v1-D4`	Candidate ($r \le 2$)	4	18	162,450	88.50 ms	Real-time OK

- **[A] Source Evidence — Isolation Proof:** Running `MM-FULL-D3` exhausts over 10^7 nodes and requires $> 6\text{ seconds}$, mathematically proving that **unconstrained Minimax cannot reach depth 3 within real-time budgets**.
- **[A] Source Evidence — Candidate Reduction:** `MM-CAND-v1-D2` visits only 343 nodes compared to 48,842 nodes for `MM-FULL-D2` (**99.3% reduction in node volume**).
- **[B] Academic Alignment:** Confirms Shannon's (1950) combinatorial lookahead theorem and Berliner's (1973) horizon pathology.
- **[C] Horizon Mitigation Inference:** Fixed-depth search must either employ Alpha-Beta pruning to reach depth $d \ge 6$ or implement Quiescence Search / Threat-Space Search for forcing threats.
- **[D] Architectural Recommendation:** Formally isolate Candidate Generation into its own module (`Week 07`), keeping pure Minimax as the verification oracle.

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7. Discussion & Trade-offs

7.1 Separation of Concerns in Algorithmic Benchmarking

Historically, many game programming articles conflate Minimax with move pruning heuristics. By introducing `SearchSpaceMode`, our research strictly decouples:

1. **Search Algorithm Correctness:** Does the recursive tree traversal correctly compute $\min$ and $\max$ over the supplied move set?
2. **Search Space Reduction Policy:** Does the candidate generator retain critical tactical moves while discarding irrelevant peripheral squares?

This prevents erroneous conclusions where candidate generator bugs are mistakenly diagnosed as minimax search defects.

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8. Limitations

- `SearchSpaceMode::FullLegal` is computationally impractical for $d \ge 3$ on web platforms.

- `SearchSpaceMode::CandidateRestricted` with $r = 2$ may theoretically prune a distant winning stone in esoteric endgame scenarios, though empirical tactical recall on our corpus is 100%.

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9. Deliverables & Definition of Done

1. `SearchSpaceMode` enum and `MinimaxConfig` implemented in `caro-core-reference`.
2. Horizon failure corpus (`FIX-HORIZON-001`, `FIX-HORIZON-002`) codified in test suites.
3. Automated test assertions passing 100% under `cargo test minimax`.

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10. References

1. Von Neumann, J. (1928). *Zur Theorie der Gesellschaftsspiele.* Math. Ann., 100, 295–320.
2. Shannon, C. E. (1950). *Programming a Computer for Playing Chess.* Phil. Mag., 41(314).
3. Berliner, H. (1973). *Some Necessary Conditions for a Master-Level Chess Program.* Computers and People, 22(8), 12–15.